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Equivalent Pain and Opioid Use Between Transabdominal and Retroperitoneal Adrenalectomy

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ABSTRACT

Introduction: Laparoscopic transabdominal adrenalectomy (LTA) and posterior retroperitoneoscopic adrenalectomy (PRA) are safe, effective surgical approaches. A direct comparison of postoperative pain and narcotic use is needed.

Methods: Adults who had laparoscopic adrenalectomy at a tertiary institution from 2015 to 2021 were identified from a surgical database. Evaluated key outcomes included opioid use and patient-reported pain during the perioperative period, which were analyzed by surgical approach and compared using multivariate linear regression. Additional data on pain postdischarge, prescription refills, length of stay, and 30-d readmission were also collected.

Results: Eighty-eight (69.3%) surgeries were LTA and 39 (30.7%) were PRA. The studied patient population was 58% female and had an average age of 54 (standard deviation 13). Adrenal tumors were 75% functional and 3.4 cm on average (standard deviation 2.6). Postoperative pain scores at rest (4.4 LTA versus 4.5 PRA, $P = 0.87$) and activity (4.7 LTA versus 5.6 PRA, $P = 0.08$) did not reach statistical significance. Patients undergoing LTA used a median of 110.3 morphine milligram equivalents (interquartile range 70.1–144.5) of opioids during their hospital stay compared to 91.0 (interquartile range 59.1–133.3) for PRA ($P = 0.16$). Linear regression demonstrated no significant difference in postoperative opioid use between approaches (−9.3 morphine milligram equivalents [95% confidence interval −40.7 to 22.1]).

Conclusions: Our analysis found no significant advantage of PRA over LTA in terms of postoperative pain or opioid use, contrary to earlier findings by Barczyński et al., who reported lower postoperative pain with PRA. Both approaches show similar clinical outcomes, and the choice between them should be based on individual patient factors rather than differences in patient-reported pain and opioid usage.

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Introduction

Over the past 100-plus years, the surgical approach for adrenal resection has evolved dramatically. The first successful adrenalectomy was reported in 1890 by J. Thornton, who described a large T-shaped incision used to remove a 20 pound adrenal tumor.¹ Many years later, in 1932, Lennox Broster performed a unilateral adrenalectomy via a posterior intercostal approach.¹ By 1992, Gagner^{1,2} and Higashihara³ described the first laparoscopic approaches for adrenalectomy, marking significant advancements in surgical technique, outcomes, and patient experience.

Today, laparoscopic adrenalectomies have become standard. The first laparoscopic adrenalectomies were performed via a laparoscopic transabdominal adrenalectomy (LTA) in a lateral decubitus position; however, by 1995, Mercan *et al.* described the first laparoscopic posterior retroperitoneoscopic adrenalectomy (PRA).¹ This approach soon gained popularity following Walz's description of a technique modification using higher insufflation pressures with improved working space.¹ Walz had great success and reported a series of 560 procedures, with 0% mortality, 1.3% major and 14.4% minor complication rates, and a conversion rate of 1.9%.⁴ Since then, both LTA and PRA have become accepted, safe, and effective surgical options for adrenalectomy.^{1,4-10}

Previous studies have shown the two approaches have equivalent clinical outcomes. Studies comparing operative time, short and long-term complications, and length of stay have demonstrated similar overall results.^{1,4,6-10} One proposed benefit of PRA is an improved patient experience through reduced postoperative pain. A previous prospective randomized controlled study of a single institution in Poland by Barczynski *et al.* suggested that PRA have lower pain scores than LTA.⁵ However, the impact of surgical approach on postoperative pain and, importantly, the amount of opioid analgesia utilized has not been evaluated in their or other similar studies. Due to the ongoing opioid epidemic, it is critical to investigate if a correlation can be drawn between surgical approach and pain and narcotic use. This study compares postoperative pain and the use of narcotics in LTA versus PRA approaches. We predicted that our retrospective data analysis would show lower pain scores and subsequent opioid associated with the PRA rather than the LTA approach.

Material and Methods

Data source

All adults (≥ 18 ys) undergoing laparoscopic adrenalectomy by the endocrine surgery or surgical oncology services at a tertiary academic institution between 2015 and 2021 were identified via a prospectively collected surgical database. Patients were excluded if they underwent an open operation (20), had surgery at a partner site (9), were pediatric patients (1), or if their adrenalectomy was combined with another procedure (4). Additionally, 8 operations were converted to open and were excluded from our primary outcome analysis. Retrospective chart review was performed by trained research staff.

A series of charts were reviewed collectively by the abstraction team, and then 10 charts were reviewed by each abstractor to confirm interuser reliability.

Data variables

Our primary outcomes of interest were the amount of opioid analgesia used and the level of patient-reported pain in the perioperative period. Opioid type, dosage, and route were abstracted from the medication administration records and converted into morphine milligram equivalents (MME). For reference, one 5 mg dose of oxycodone is equivalent to 7.5 MME. Intraoperative analgesia given was obtained from a review of anesthesia records.

At our institution, nursing staff routinely obtain and record patient pain scores as part of their patient rounds. Most are measured using the numerical rating scale (NRS), a measure of self-reported pain using a scale from 0 ("no pain") to 10 ("worst pain imaginable"). NRS scores at rest and with movement were recorded when available. Scores were taken throughout the day and averaged. The average opioid use and pain scores were calculated by operative day, with postoperative day 0 considered the day of surgery until midnight. Subsequent postoperative days were counted, starting at midnight to the following midnight. Particular attention was paid to the immediate postoperative period, as 81% of patients were discharged by postoperative day 1.

Additional postoperative information was captured for each surgical approach, including if the patient mentioned significant pain in the 2 wks between discharge and the postoperative visit and if an opioid prescription was refilled. Patients had regular follow-ups in the clinic 2 wks postoperatively and then as needed. Pain was assessed during the postoperative visits, and this information was abstracted from the clinic notes. Opioid prescription refill data was also abstracted from clinic notes.

Secondary outcomes evaluated included length of stay and 30-d readmission. Surgical complications were also documented and categorized as capsular disruption, visceral injury, or hemorrhage requiring transfusion.

Patient demographics were abstracted, including age, sex, race/ethnicity, weight, history of prior abdominal surgery, and comorbidities. Basic tumor information, including maximum tumor size on final pathology, surgical indication (functional adenoma, nonfunctional adenoma, or malignancy [primary adrenal or metastatic]), and laterality, was also abstracted.

Surgical approaches

Most surgeons included during this period performed both laparoscopic transabdominal and retroperitoneal adrenalectomy. Transabdominal adrenalectomy was performed in the lateral decubitus position and typically utilized 3-4 ports. Retroperitoneal adrenalectomy was performed in the prone, jackknife position and commonly used 3 ports. The choice of approach was based on surgeon preference, owing primarily to patient and tumor anatomy. In general, LTA was utilized preferentially when the adrenal gland was more anterior to the kidney. While body mass index (BMI) or weight is not a

specific consideration, fat distribution in the hips or retroperitoneum makes RPA less favored compared to intra-abdominal fat distribution, making LTA less favored. The operative approach is also influenced by a history of previous abdominal surgeries or back/flank procedures. Finally, LTA is often preferred in operations for larger tumors, as dissection and tumor mobilization can be more challenging via the posterior approach. Robotic adrenalectomy was not performed at our institution during this period. All patients were routinely observed for at least 1 night postoperatively.

Analgesics utilized

While local anesthetic was used in each case, other adjuncts, such as transabdominal plane blocks, epidural anesthesia, or liposomal bupivacaine, were not. The amount and type of intraoperative analgesia provided were at the discretion of the anesthesiologist. A general, standardized postoperative order set that includes options for multimodal analgesia was used. As a result, almost all patients (93.7%) received acetaminophen postoperatively. Ketorolac and other nonsteroidal anti-inflammatory drugs, which are part of the order set, were ordered at the discretion of the operating surgeon. Other adjuncts, such as muscle relaxants and gabapentin, are not part of the order set and are not routinely used (data on their usage was not recorded). No changes were made to this generic order set before or during the study period. There was no set protocol for the number of opioids patients were discharged with.

Statistical analysis

Comparisons of categorical variables were evaluated using chi-squared analysis and continuous variables using a student's t-test. Normality was tested using the Shapiro–Wilk test. The Wilcoxon rank sum test was used to measure data that did not meet the normality criteria. Multivariable linear

regression was performed, looking at the association of surgical approach with our primary outcomes while controlling for factors potentially contributing to patient pain and analgesia use, including patient age, sex, weight, BMI, tumor size, use of ketorolac, and laterality. All analyses were performed using SAS (v9.4, Cary NC). The Institutional Review Board of the University of Wisconsin approved the study and deemed it exempt from review.

Results

Demographics

Between 2015 and 2021, there was a total of 169 patients who underwent adrenalectomy, of whom 127 met inclusion criteria. Overall, the study population had a mean age of 54.4 ys (standard deviation 13.3) and was 57.5% female. The average tumor size was 3.5 cm (standard deviation 2.4 cm); 75.6% of tumors were functional, 19.7% were nonfunctional adenomas, and 4.7% were malignancies. In all, 55.9% of tumors were left-sided, compared to 43.3% right-sided. There was one bilateral operation.

Of the 127 operations, 88 (69.3%) were performed via the transabdominal approach and 39 (30.7%) were performed by the retroperitoneal approach. Patient demographics were similar between groups (Table 1); the one notable difference was that patients undergoing LTA had a higher percentage of female patients (64.8%) compared to PRA (41.0%, $P = 0.02$). Patient BMI was higher in the LTA group (mean 34.0, standard deviation 8.8) compared with the PRA group (mean 30.7, standard deviation 5.1) ($P = 0.03$) (Table 1). On average, patients undergoing LTA had slightly larger tumors (4.0 cm versus 2.3 cm, $P < 0.01$). There was no significant difference in tumor pathology or laterality (Table 2). The proportion of PRA and LTA was similar throughout the study (Table 3).

Table 1 – Patient demographics by surgical approach.

Characteristic	Laparoscopic transabdominal adrenalectomy		Posterior retroperitoneoscopic adrenalectomy		P value
	n	%	n	%	
Age (mean, SD)	55.0	13.2	53.1	13.4	0.44
Weight, lbs. (mean, SD)	214.6	56.0	198.9	55.3	0.15
BMI (mean, SD)	34.1	8.8	30.7	5.1	0.03
Gender					0.02
Female	57	64.8	16	41.0	
Male	31	35.2	22	56.4	
Comorbidities					0.42
0	9	10.2	4	10.3	
1	32	36.4	13	33.3	
2	27	30.7	8	20.5	
3+	20	22.7	14	35.9	
History of abdominal surgery	28	31.8	13	33.3	0.87
Total	88	69.3	39	30.7	

Table 2 – Comparison of tumor characteristics by surgical approach.

Tumor characteristic	Laparoscopic transabdominal adrenalectomy		Posterior retroperitoneoscopic adrenalectomy		P value
	n	%	n	%	
Indication					0.43
Functional	64	72.7	32	82.1	
Nonfunctional	20	22.7	5	12.8	
Cancer	4	4.6	2	5.1	
Tumor size					<0.01
Mean	4.0	2.6	2.3	1.8	
<1 cm	7	8.0	9	23.1	
1-4 cm	42	47.7	24	61.5	
>4 cm	39	44.3	6	15.4	
Side					0.12
Left	54	61.4	17	43.6	
Right	33	37.5	22	56.4	
Bilateral	1	1.1	0	0	
Total	88	69.3	39	30.7	

Pain scores

Both groups' pain scores at rest were relatively similar (Table 4). On postoperative day 0, pain at rest was rated an average of 4.4 for LTA and 4.5 for PRA ($P = 0.87$). By postoperative day 1, average pain at rest had decreased to 4.0 for LTA and 3.5 for PRA ($P = 0.17$). When looking at pain with activity on postoperative day 0, PRA trended toward slightly higher scores at 5.6 compared to 4.7 ($P = 0.08$). By postoperative day 1, average pain with activity for PRA decreased to 4.7, while it increased to 5.2 for LTA; the difference in average score was not statistically significant ($P = 0.24$). Postoperative pain reports were similar at 2 wks, regardless of approach (Table 5). 31.8% of patients who underwent LTA reported pain compared to 33.3% who underwent PRA ($P = 0.55$). However, this data is limited as 21.3% of surgeons' notes did not mention pain in an affirmative or negative manner.

Opioid utilized

The overall amount of perioperative opioids used was also similar between groups (Table 6). Patients undergoing LTA

used a median of 110.3 MME (IQR 70.1-144.5) of opioids during their hospital stay compared to 91.0 (IQR 59.1-133.3) for PRA ($P = 0.16$). When looking at more granular time portions of the perioperative period – intraoperatively, postoperative day 0, and postoperative day 1 – there was also no significant difference in the amount of opioids used (Table 4). Further, there was no significant difference in the percentage of patients receiving IV versus PO opioids between the two groups (Table 6).

At 2 wks postoperatively, there was no statistically significant difference in opioid prescription refills by approach (Table 5). As abstracted from clinic notes, 14.8% of LTA and 12.8% of PRA patients obtained refills ($P = 0.55$).

Other analgesics utilized

Nearly all patients received perioperative acetaminophen, and there was no difference in receipt between approaches (Table 6). There was a significantly higher percentage of patients undergoing LTA who received ketorolac (19.3%) compared to those undergoing PRA (5.1%, $P = 0.04$).

Table 3 – Number of cases by surgical approach by year with annual percentages.

Approach	Year								
	2015	2016	2017	2018	2019	2020	2021	2022	Total
LTA	15	10	12	8	11	7	16	9	88
	93.75	76.92	54.55	42.11	68.75	77.78	76.19	81.82	69.29%
PRA	1	3	10	11	5	2	5	2	39
	6.25	23.08	45.45	57.89	31.25	22.22	23.81	18.18	30.71%
Total	16	13	22	19	16	9	21	11	127
	12.60	10.24	17.32	14.96	12.60	7.09	16.54	8.66	100.00%

Table 4 – Average numerical rating system pain scores by surgical approach.

Characteristic	Laparoscopic transabdominal adrenalectomy		Posterior retroperitoneoscopic adrenalectomy		P value
	Mean	SD	Mean	SD	
Average POD0 pain score at rest	4.4	2.2	4.5	1.7	0.87
Average POD0 pain score with activity	4.7	2.4	5.6	2.2	0.08
Average POD1 pain score at rest	4.0	2.1	3.5	2.0	0.17
Average POD1 pain score with activity	5.2	1.9	4.7	2.1	0.24

POD = postoperative day.

Linear regression

Multivariable linear regression evaluating factors associated with total opioid received demonstrated that patients undergoing PRA had a slightly lower MME, but this was not statistically significant (*versus* LTA -9.3 MME [95% confidence interval -40.7 to 22.1]). As the size of tumors increased, so did the amount of opioid used (8.9 MME per cm [95% confidence interval 3.2 - 14.6]). There was no significant association with patient age, weight, or tumor laterality. Female patients had lower opioid use (*versus* male -30.2 MME [95% confidence interval -57.9 to -2.4]) (Table 7).

Clinical outcomes

Overall, the average length of stay was 1.3 ds (standard deviation 0.6) (Table 8). Most patients (81.1%) were discharged on postoperative day 1, with another 14.2% discharged on postoperative day 2. There was no difference in the average length of stay between approaches (1.3 ds for LTA *versus* 1.2 ds for PRA, $P = 0.29$). 87.2% of patients undergoing PRA were discharged on postoperative day 1 compared to 78.4% ($P = 0.48$); 95.5% and 94.9% of patients undergoing LTA and PRA, respectively, were discharged by postoperative day 2.

Surgical complication rates (including capsular disruption, visceral injury, and hemorrhage requiring transfusion) were generally low, with only 14 cases having an identified

complication (11%). There were two visceral injuries, both in LTA cases, which were repaired laparoscopically. Two patients required transfusion, one in LTA and one in PRA. There was no statistical difference in complication rates between approaches (all $P > 0.05$, Table 2). Similarly, there was no difference between approaches for postoperative emergency department visits or 30-d readmission rates (Table 8). Although excluded from our primary analysis, a total of 8 of 135 eligible cases were converted to open (5.9%); 6 were done using the LTA approach (6.4%), while 2 were performed using PRA (4.9%, $P = 0.73$).

Discussion

Our data analysis suggests no superiority of the PRA over the LTA approach when comparing postoperative pain or opioid use in the immediate postoperative period. This contrasts the prior study by Barczyński *et al.* that concluded that PRA had decreased pain intensity and analgesic use postoperatively compared to LTA.⁵ Notably, the Barczyński study assessed pain with the Visual Analog Scale (VAS), while our study used the NRS. Given the use of two different pain scales, directly comparing patients' pain experiences is difficult. The exact reason for the difference in findings is unclear, especially as the surgical techniques used for LTA and PRA were the same in both studies. One possible explanation is that our patients

Table 5 – Postoperative opioid refills and pain reports by surgical approach.

Characteristic	Laparoscopic transabdominal adrenalectomy		Posterior retroperitoneoscopic adrenalectomy		P value
	n	%	n	%	
Postoperative opioid refill					0.55
Yes	13	14.8	5	12.8	
No	60	68.2	30	76.9	
Missing	15	17.1	4	10.3	
Postoperative pain reported in note					0.55
Yes	28	31.8	13	33.3	
No	39	44.3	20	51.3	
Missing	21	23.9	6	15.4	

Table 6 – Analgesic use by surgical approach.

Characteristic	Laparoscopic transabdominal adrenalectomy		Posterior retroperitoneoscopic adrenalectomy		P value
	Median	IQR	Median	IQR	
Average total perioperative opioid use (MME)	110.3	70.1-144.5	91.0	59.1-133.3	0.16
Average intraoperative opioid use (MME)	54.4	32.20-73.5	54.8	30.5-68.0	0.57
Average POD0 opioid use (MME)	26.5	12.8-42.5	24.3	11.6-35.2	0.43
Average POD1 opioid use (MME)	30.0	15.0-55.0	21.3	15.0-45.0	0.22
Received IV opioid on POD0 (% received)	70.3		29.7		0.85
Received IV opioid on POD1 (% received)	63.0		37.0		0.85
Postoperative acetaminophen (% received)	92.1		97.4		0.25
Postoperative ketorolac (% received)	19.3		5.1		0.04

IQR = interquartile range; POD = postoperative day.

undergoing LTA were more likely to receive ketorolac postoperatively, which could lower patient-reported pain scores of LTA compared to PRA. Our adjusted model showed no statistically significant association between ketorolac use and reported pain and opioid use, but it did trend toward lower opioid use. Additionally, while tumor size was similar in both studies, our patients had a larger BMI on average (33 versus 27), which could contribute to the amount of dissection needed and pain experienced. Finally, it should be noted that the Barczyński study was performed in Poland in the early 2000s compared to our current review of American patients. Cultural factors influence perceptions and reports of pain and may contribute to some of the differences we found.¹¹ Additionally, the use and expectation of opioids for analgesia vary significantly across the globe, with the United States far outstripping other countries, including Poland.¹²

Several other factors were positively associated with increased postoperative opioid use after surgery. One of the most significant factors we identified was tumor size, most likely due to the larger incision size and potentially increased operative time due to complexity. Using analgesic adjuncts, for example, nerve blocks or multimodal pain control

approaches, might reduce opioid necessity for patients with larger tumors, and the cost and benefit of such adjuncts warrant further exploration in adrenal surgery. Additionally, this study data suggests that female patients used fewer MME following laparoscopic adrenalectomy than male patients. Studies such as the ones by Hah *et al.* and Hrebinko *et al.* have evaluated patient characteristics and identified a patient's sex as being associated with pain scores, opioid use, recovery, and opioid cessation in the perioperative period.^{13,14} However, there is a lack of literature regarding sex-based characteristics and their impact on narcotic use following adrenal surgery by any approach.

Finally, contradictory literature reports variance in key clinical outcomes of adrenalectomy by approach. Overall, it should be noted the safety and efficacy of both approaches are excellent.^{15,16} Studies comparing the two have not determined a definitive consensus, partially attributable to varied selection criteria for each approach. Dickinson *et al.* found that when treating pheochromocytomas, PRA, compared to LTA, accomplished reduced operative times, estimated blood loss, and LOS.⁸ Given careful selection criteria, Berber *et al.* concluded that each approach achieved similar perioperative outcomes. They expounded that PRA was favored in patients with previous abdominal surgeries, smaller tumors (<6 cm), lower BMI patients, and those with aldosteronomas, Cushing's, and bilateral pathologies.⁶ Our study found no differences in clinical outcomes of adrenalectomy by approach. Specifically, length of stay, complication rates, postoperative emergency department visits, and 30-d readmission rates were comparable and low. This further supports the idea that both approaches are safe and valuable for the contemporary endocrine surgeon.

Single-institution data limit this study, and the results may not be broadly applicable. This study did not randomly assign patients to each treatment group; therefore, selection bias could impact results. Additionally, information related to chronic pain or patients' opioid use before the start of this study could not be reliably abstracted during data collection. This is a potential study limitation and could result in bias. Immediate postoperative outpatient opioid use was evaluated, and 2-wk postoperative data suggest similar pain outcomes by approach. However, this data is incomplete and fails

Table 7 – Adjusted linear regression of amount of opioid utilized (MME).

Variable	Beta	95% CI	
Intercept	63.4	–32.8	159.6
Retroperitoneal	–9.3	–40.7	22.1
Age (y)	–1.1	–2.1	0.0
Patient weight (lbs.)	0.1	–0.1	0.3
Female	–30.2	–57.9	–2.4
Tumor size (cm)	8.9	3.2	14.6
Left-sided tumor	–0.8	–28.0	26.3
Ketorolac use	–26.6	–65.7	12.5
Complication*	34.0	–10.7	78.7

*Complications included visceral injury, capsular disruption, or need for transfusion.

Table 8 – Surgical outcomes by surgical approach.

Characteristic	Laparoscopic transabdominal adrenalectomy		Posterior retroperitoneoscopic adrenalectomy		P value
	n	%	n	%	
Capsular disruption	5	5.7	5	12.8	0.17
Visceral injury	2	2.3	0	0	0.34
Hemorrhage requiring transfusion	1	1.1	1	2.6	0.55
Length of stay					0.48
1 d	69	78.4	34	87.2	
2 d	15	17.1	3	7.7	
3+ d	4	4.5	2	5.1	
Postoperative ED visit	5	5.7	2	5.1	0.90
30-d readmission	2	2.3	2	5.1	0.40
Total	88	69.3	39	30.7	

ED = emergency department.

to capture long-term pain outcomes, as a significant number of encounters did not mention pain in an affirmative or negative manner. Though patients were not discharged from the hospital with a standardized amount of opioids, data shows there was no significant difference in refill rates between approaches; specific dosing or amounts at discharge or obtained via refill are not known. Reports of the development of chronic pain and the need for longer-duration opioid analgesia (dose, amount, and duration) in each treatment group are a potential focus of future research. Finally, pain levels were reported as a cumulative daily pain score and with the number of recordings per day rather than as discrete pain values. Though this NRS data collection allowed for continuous rather than ordinal values, the values were normally distributed, as evidenced by the Shapiro–Wilk test, and therefore met our inclusion criteria.

Despite study limitations, these results show that both transabdominal or retroperitoneal approaches for adrenalectomy are safe and have similar patient-centered outcomes of reported pain and need for opioid analgesia. As the opioid epidemic continues to impact the United States significantly, surgeons should be mindful of how surgical techniques influence pain and opioid use. Since our study demonstrated no significant difference in patient experience or opioid use, this, fortunately, need not be considered when deciding on PRA versus LTA for a patient. Instead, joint decision-making regarding the operative approach should be based on specific clinical circumstances.

Conclusions

This study demonstrated that PRA and LTA have similar pain scores and postoperative opioid use, suggesting equivalency between approaches in these patient-centric metrics.

Disclosure

The authors declare that there are no conflicts of interest for this work. We confirm that this work is original and has not

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Lauren Haskins: Writing – review & editing, Writing – original draft, Project administration, Formal analysis. **Nicole Tobin:** Writing – review & editing, Project administration, Methodology, Data curation. **Elle Reineman:** Project administration, Data curation. **Rebecca Sippel:** Writing – review & editing, Methodology. **David Schneider:** Writing – review & editing, Methodology. **Kristin Long:** Writing – review & editing, Methodology. **Dawn Elfenbein:** Writing – review & editing, Methodology. **Courtney Balentine:** Writing – review & editing, Methodology. **Alexander Chiu:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

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